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Control device

Abstract

In the case where drive power of a vehicle is increased, a control device starts micro-braking control for generating a braking force on at least one wheel of a front wheel and a rear wheel. During execution of the micro-braking control, the control device estimates an efficacy factor that is a factor indicating a relationship of a magnitude of the braking force to a magnitude of a pressing force generated by a friction braking system. The control device estimates the efficacy factor for the wheel, to which the braking force is applied by the micro-braking control, on the basis of a deviation between generated drive power, which is based on a required value for a power source, and actual drive power actually acting on the vehicle, and the pressing force corresponding to the wheel, to which the braking force is applied by the micro-braking control.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a control device that is applied to a vehicle including a friction braking system.

BACKGROUND ART

(2) PTL 1 discloses a braking control device that estimates, as a degree of braking effectiveness, a relationship indicating a magnitude of an actual braking force to a wheel cylinder pressure when a brake pedal is depressed. More specifically, it is configured to actuate a brake and estimate the degree of the braking effectiveness on the basis of a deviation between actual deceleration and estimated deceleration, the actual deceleration being calculated from front-rear acceleration that is based on a detection signal from a front-rear acceleration sensor, and the estimated deceleration being calculated from the wheel cylinder pressure.

CITATION LIST

Patent Literature

(3) PTL 1: JP2007-196705A

SUMMARY

Technical Problem

(4) In some cases, a relationship between a magnitude of a pressing force to press a friction member against a rotary body integrally rotating with a wheel and a magnitude of a braking force applied to the wheel differs between a front wheel and a rear wheel of vehicle wheels. The braking control device disclosed in PTL 1 uniformly estimates the degree of the braking effectiveness and does not take the difference between the front wheel and the rear wheel into consideration.

Solution to Problem

(5) In the summary, a control device for solving the above problem is a control device that is applied to a vehicle having: a power source that generates drive power; and a friction braking system that applies a braking force to a wheel according to a pressing force to press a friction member against a rotary body integrally rotating with the wheel and can adjust the braking forces applied to a front wheel and a rear wheel of the wheels separately, and includes: a braking control unit that controls the friction braking system; and an estimation unit that estimates an efficacy factor as a factor indicating a relationship of a magnitude of the braking force to a magnitude of the pressing force. The braking control unit can start micro-braking control for generating the braking force on at least one wheel of the front wheel and the rear wheel in the case where the drive power is increased. During execution of the micro-braking control, the estimation unit estimates the efficacy factor for the wheel, to which the braking force is applied by the micro-braking control, on the basis of a deviation between generated drive power that is based on a required value for the power source and actual drive power that actually acts on the vehicle, and the pressing force that corresponds to the wheel, to which the braking force is applied by the micro-braking control.

(6) In the above configuration, the micro-braking control is executed as control for applying the braking force in order to estimate the efficacy factor. In the micro-braking control, the braking force is generated on at least one wheel of the front wheel and the rear wheel. Then, the pressing force corresponding to the wheel, to which the braking force is applied, is used to estimate the efficacy factor. In this way, the efficacy factor for the front wheel and the efficacy factor for the rear wheel can be estimated distinctively.

(7) Furthermore, the micro-braking control is started in the case where the drive power is increased. The micro-braking control is executed in the case where the drive power is increased. In this way, even when the braking force is generated on at least one wheel of the front wheel and the rear wheel, it is possible to alleviate an impact of generation of the braking force, such as an occupant of the vehicle having a sense of discomfort and pitching motion of the vehicle.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view illustrating an embodiment of a control device for a vehicle and a vehicle including the control device.

(2) FIGS. 2A and 2B are timing charts exemplifying a relationship between transitions of a braking hydraulic pressure and drive power in the case where a braking force is applied during an increase in the drive power.

(3) FIG. 3 is a graph illustrating a relationship among the braking hydraulic pressure, an efficacy factor, and a reduced amount of the drive power.

(4) FIG. 4 is a flowchart illustrating a processing flow at the time when the control device executes micro-braking control.

(5) FIG. 5 is a graph illustrating a map for calculating a required braking force during execution of the micro-braking control.

(6) FIG. 6 is a flowchart illustrating a processing flow at the time when the control device estimates

the efficacy factor.

(7) FIG. 7 is a block diagram illustrating an efficacy factor estimation unit.

(8) FIGS. 8A and 8B are timing charts illustrating a transition of the braking force by the micro-braking control that is executed by the control device.

(9) FIGS. 9A, 9B and 9C are timing charts illustrating a transition of a braking force by micro-braking control that is executed by a control device in a modified example.

DESCRIPTION OF EMBODIMENTS

(10) A description will hereinafter be made on an embodiment of a control device with reference to FIG. 1 to FIG. 8A.

(11) FIG. 1 illustrates a vehicle 90 that includes a power source 91, a friction braking system 70, and a control device 10. A control target of the control device 10 is the friction braking system 70. The friction braking system 70 can generate a frictional braking force to the vehicle 90.

(12) FIG. 1 illustrates one of front wheels 81 of wheels provided to the vehicle 90 and one of rear wheels 82 of the wheels provided to the vehicle 90.

(13) The vehicle 90 includes a braking operation member 79. The braking operation member 79 can be operated by a driver of the vehicle 90. An example of the braking operation member 79 is a brake pedal.

(14) The vehicle 90 may include an autonomous driving control device 30 that calculates a command value for autonomous travel of the vehicle 90. The autonomous driving control device 30 can exchange information with the control device 10.

(15) <Power Source>

(16) An example of the power source 91 provided to the vehicle 90 is an electric motor. The power source 91 is not limited to the electric motor and may be an internal combustion engine.

Alternatively, the electric motor and the internal combustion engine may each be adopted as the power source 91. Further alternatively, the power source 91 may be an in-wheel motor that is the electric motor attached to a wheel of each of the wheels of the vehicle 90.

(17) <Friction Braking System>

(18) A description will be made on the friction braking system 70. The friction braking system 70 includes braking mechanisms 73F, 73R, each of which corresponds to respective one of the wheels of the vehicle 90. FIG. 1 exemplifies the braking mechanism 73F corresponding to the front wheel 81 and the braking mechanism 73R corresponding to the rear wheel 82.

(19) An example of the friction braking system 70 is a hydraulic braking system. The friction braking system 70 as the hydraulic braking system includes: a hydraulic pressure generator 71; and a braking actuator 72 to which a brake fluid is supplied from the hydraulic pressure generator 71. The braking operation member 79 is coupled to the hydraulic pressure generator 71. The hydraulic pressure generator 71 can generate a hydraulic pressure that corresponds to a braking operation amount as an operation amount of the braking operation member 79 by the driver of the vehicle 90. The hydraulic pressure generator 71 can also generate the hydraulic pressure on the basis of the command value that is calculated by the autonomous driving control device 30.

(20) Since components are common between the braking mechanisms 73F, 73R, the components are denoted by common reference signs for description. The braking mechanisms 73F, 73R are each configured to include: a wheel cylinder 74; a rotary body 76 that rotates integrally with the wheel; and a friction member 75 that can be pressed against the rotary body 76. An example of each of the braking mechanisms 73F, 73R is a disc brake. Each of the braking mechanisms 73F, 73R may be a drum brake.

(21) The braking actuator 72 is connected to each of the wheel cylinders 74. For example, when the braking operation member 79 is operated, the brake fluid in an amount corresponding to the operation amount thereof is supplied to each of the wheel cylinders 74. The hydraulic braking system can generate the frictional braking force according to a WC pressure that is the hydraulic pressure in each of the wheel cylinders 74 provided to the braking mechanisms 73F, 73R. The

braking mechanisms **73F**, **73R** are each configured that, as the WC pressure is increased, a force of pressing the friction member **75** against the rotary body **76** integrally rotating with the wheel is increased. Each of the braking mechanisms **73F**, **73R** can apply the larger braking force to the wheel as the WC pressure is increased. The WC pressure is an example of a value that indicates a pressing force to press the friction member **75** against the rotary body **76**.

(22) The braking actuator **72** can change the hydraulic pressure in each of the wheel cylinders **74** separately such that a magnitude of the braking force applied to each of the wheels differs. A sum of the braking forces that are applied to the front wheels **81** by the braking mechanisms **73F** provided in a manner to correspond to the front wheels **81** will be referred to as a front-wheel braking force. A sum of the braking forces that are applied to the rear wheels **82** by the braking mechanisms **73R** provided in a manner to correspond to the rear wheels **82** will be referred to as a rear-wheel braking force.

(23) <Sensor>

(24) The vehicle **90** includes various sensors. FIG. **1** illustrates, as an example of the various sensors, a hydraulic pressure sensor **61**, a front-rear acceleration sensor **62**, and a wheel speed sensor **63**. Detection signals from the various sensors are input to the control device **10**.

(25) The hydraulic pressure sensor **61** is a sensor that detects the pressing force for pressing the friction member **75** against the rotary body **76** in the friction braking system **70**. As an example, the hydraulic pressure sensor **61** can detect the WC pressure in each of the wheel cylinders **74** as a braking hydraulic pressure P . In this case, the hydraulic pressure sensor **61** is attached in a manner to correspond to each of the wheel cylinders **74**. A value that is detected as the braking hydraulic pressure P is not limited to the WC pressure as long as such a value is a pressure corresponding to the pressing force.

(26) The front-rear acceleration sensor **62** is a sensor that detects acceleration in a front-rear direction of the vehicle **90**.

(27) The wheel speed sensor **63** is a sensor that detects a wheel speed. The wheel speed sensor **63** is provided to each of the wheels. A vehicle speed can be calculated on the basis of the wheel speed.

(28) <Control Device>

(29) A description will be made on the control device **10**. The control device **10** is configured to include plural functional units that execute various types of control. FIG. **1** illustrates, as an example of the functional units, a drive control unit **11**, a braking control unit **12**, an acquisition unit **13**, a friction member temperature estimation unit **14**, a storage unit **15**, and an efficacy factor estimation unit **20**. The efficacy factor estimation unit **20** corresponds to an estimation unit that estimates an efficacy factor K . The functional units that are provided to the control device **10** can exchange information with each other.

(30) The vehicle **90** may include another control device instead of the control device **10**. In addition, some of the functional units provided to the control device **10** may be provided to the other control device.

(31) Here, the control device **10**, the autonomous driving control device **30**, and the other control device may each have any of the following configurations [a] to [c]. [a] One or more processors that execute various types of processing according to a computer program are provided. The processor includes a processing unit. Examples of the processing unit are a CPU, a DSP, and a GPU. The processor includes memory. Examples of the memory include RAM, ROM, and flash memory. The memory stores a program code or a command configured to cause the processing unit to execute the processing. The memory, that is, a computer-readable medium includes any available medium that can be accessed by a general-purpose or dedicated computer. [b] One or more hardware circuits that execute the various types of the processing are provided. Examples of the hardware circuit are an Application Specific Integrated Circuit (ASIC), a Complex Programmable Logic Device (CPLD) and a Field Programmable Gate Array (FPGA). [c] A circuit including: the processor that partially executes the various types of the processing according to the computer

program; and the hardware circuit that executes rest of the various types of the processing are provided.

(32) A description will be made on the drive control unit **11**. The drive control unit **11** can control the power source **91**. The drive control unit **11** can actuate the power source **91** and transmit drive power to a drive wheel. For example, the drive control unit **11** calculates a required value for the power source **91** that corresponds to an operation of an accelerator operation member by the driver of the vehicle **90**, and thus can transmit the drive power from the power source **91** on the basis of the required value. Alternatively, for example, the drive control unit **11** can transmit the drive power from the power source **91** on the basis of the command value that is calculated by the autonomous driving control device **30**. The drive control unit **11** can calculate generated drive power F_p as the drive power, which is output by the power source **91**, on the basis of the required value for the power source **91**. For example, the generated drive power F_p can be calculated on the basis of a workload of the electric motor that is mounted as the power source **91**.

(33) A description will be made on the braking control unit **12**. The braking control unit **12** can control the friction braking system **70**. The braking control unit **12** can actuate and cause the friction braking system **70** to generate the braking force.

(34) A description will be made on exemplary functions of the braking control unit **12**.

(35) The braking control unit **12** can calculate a required braking force BP as a required value of the braking force to be applied to the vehicle **90**. For example, the required braking force BP can be calculated on the basis of the operation amount of the braking operation member **79**. In addition, for example, the required braking force BP can be acquired on the basis of the command value that is calculated by the autonomous driving control device **30**.

(36) The braking control unit **12** can calculate a front-rear distribution ratio a as a value that indicates a ratio of the braking force applied to the front wheel **81** to the required braking force BP . The front-rear distribution ratio a indicates a ratio of the braking force applied to the front wheel **81** to a total braking force that is a sum of the braking force applied to the front wheel **81** and the braking force applied to the rear wheel **82**. The braking control unit **12** can calculate a required value of the front-wheel braking force and a required value of the rear-wheel braking force on the basis of the required braking force BP and the front-rear distribution ratio a . Hereinafter, the required value of the front-wheel braking force will be referred to as a required front-wheel braking force BP_f . In addition, the required value of the rear-wheel braking force will be referred to as a required rear-wheel braking force BP_r . For example, when the front-rear distribution ratio a is “1”, the required front-wheel braking force BP_f is equal to the required braking force BP . In addition, for example, when the front-rear distribution ratio a is “0”, the required rear-wheel braking force BP_r is equal to the required braking force BP .

(37) The braking control unit **12** can convert between the required value of the braking force and a target value of the pressing force by using a conversion factor K_b . The conversion factor K_b is a factor that indicates a relationship of the magnitude of the braking force to a magnitude of the pressing force. For example, the required front-wheel braking force BP_f can be converted to a target value of the WC pressure for the front wheel **81**. Similarly, the required rear-wheel braking force BP_r can be converted to a target value of the WC pressure for the rear wheel **82**. The braking control unit **12** actuates the friction braking system **70** such that the generated pressing force follows the target value. The braking control unit **12** stores, as an initial value of the conversion factor K_b , a value that is calculated in advance on the basis of specifications of the vehicle **90**, experimental data, and the like.

(38) The braking control unit **12** can execute micro-braking control. As will be described in detail below, the micro-braking control is control for generating the braking force on at least one wheel of the front wheel **81** and the rear wheel **82** in order to execute factor estimation processing by the efficacy factor estimation unit **20**.

(39) A description will be made on the acquisition unit **13**. The acquisition unit **13** can calculate a

state amount of the vehicle **90** on the basis of the detection signals from the various sensors. For example, the acquisition unit **13** can calculate the braking hydraulic pressure P on the basis of the detection signal from the hydraulic pressure sensor **61**. The acquisition unit **13** can calculate front-rear acceleration G_x on the basis of the detection signal from the front-rear acceleration sensor **62**. The acquisition unit **13** can calculate the wheel speed on the basis of the detection signal from the wheel speed sensor **63**.

(40) The acquisition unit **13** can calculate vehicle weight M as weight of the vehicle **90**. In addition to the weight of the vehicle **90** itself, weight of an occupant and a load is considered for the vehicle weight M . The vehicle weight M can be calculated by using the generated drive power F_p and the front-rear acceleration G_x at the time when the generated drive power F_p acts on the vehicle **90**, for example. A method for calculating the vehicle weight M is not particularly limited and may be calculated by a least-squares method, or an arithmetic mean of values calculated multiple times may be used as the vehicle weight M .

(41) The friction member temperature estimation unit **14** can estimate a temperature of the friction member **75**. The temperature of the friction member **75** can be estimated by cumulating a temperature increase amount and a temperature reduction amount, for example. The temperature increase amount can be calculated from a reduction rate of the wheel speed during braking of the vehicle **90** and the WC pressure, for example. The temperature reduction amount can be calculated on the basis of a time when the friction member **75** is not in contact with the rotary body **76**, for example.

(42) A description will be made on the storage unit **15**. The storage unit **15** can store detection values acquired from the various sensors. The storage unit **15** can also store a calculated value that is calculated by each of the functional units.

(43) A description will be made on the efficacy factor estimation unit **20**. The efficacy factor estimation unit **20** can execute the factor estimation processing to estimate the efficacy factor K . The efficacy factor K is the factor that indicates the relationship of the magnitude of the braking force to the magnitude of the pressing force. The efficacy factor estimation unit **20** executes the factor estimation processing during execution of the micro-braking control. In the factor estimation processing, the efficacy factor K is estimated for the wheel, to which the braking force is applied by the micro-braking control, on the basis of a deviation between the generated drive power F_p and actual drive power F_x that actually acts on the vehicle **90**, and the pressing force that corresponds to the wheel, to which the braking force is applied by the micro-braking control. The efficacy factor estimation unit **20** can estimate each of the efficacy factor K for the front wheel **81** and the efficacy factor K for the rear wheel **82**. As illustrated in FIG. 7, the efficacy factor estimation unit **20** includes a deviation calculation section **21**, a fluctuation pressure calculation section **22**, and an output section **23**.

(44) <Use of Efficacy Factor>

(45) The efficacy factor K that is estimated by the efficacy factor estimation unit **20** can be used as follows. For example, the efficacy factor K can be used to correct the conversion factor K_b that is stored in the braking control unit **12**. For example, the braking control unit **12** can replace the efficacy factor K as the new conversion factor K_b . For example, the braking control unit **12** can adjust the front-wheel braking force and the rear-wheel braking force by adjusting the front-rear distribution ratio a on the basis of a difference between the efficacy factor K for the front wheel **81** and the efficacy factor K for the rear wheel **82**.

(46) <Estimation Principle of Efficacy Factor>

(47) A description will be made on a principle to estimate the efficacy factor K by the factor estimation processing with reference to FIGS. 2A and 2B and FIG. 3.

(48) In an example illustrated in FIG. 2A, the drive power is increased in a period from timing t_1 to timing t_2 . In FIG. 2A, the generated drive power F_p is indicated by a solid line. In addition, the actual drive power F_x is indicated by a broken line. In the period from the timing t_1 to the timing t_2

in which the generated drive power F_p is increased, the braking force is generated. For this reason, as illustrated in FIG. 2B, the braking hydraulic pressure P is generated in the period from the timing t_1 to the timing t_2 . A specific example of the generated drive power F_p is a sum of the drive power for all the wheels transmitted from the power source **91** to wheel ground surfaces, for example. A specific example of the actual drive power F_x is a sum of the drive power for all the wheels actually generated on the wheel ground surfaces, for example.

(49) Since the braking force is applied to the vehicle **90** in the period from the timing t_1 to the timing t_2 , as indicated by the broken line in FIG. 2A, the actual drive power F_x that actually acts on the vehicle **90** is lower than the generated drive power F_p . At this time, the actual drive power F_x becomes lower than the generated drive power F_p as the braking force is increased. In other words, as the braking force is increased, a drive power deviation ΔF , which is a deviation between the generated drive power F_p and the actual drive power F_x , is increased.

(50) That is, by using the drive power deviation ΔF , which is the deviation between the generated drive power F_p and the actual drive power F_x , and a fluctuation amount of the braking hydraulic pressure P , it is possible to estimate the braking force, which actually acts on the vehicle **90**, on the basis of the drive power deviation ΔF that corresponds to the fluctuation amount of the braking hydraulic pressure P . In this way, it is possible to calculate the braking hydraulic pressure P , that is, a relationship between the pressing force and the actual braking force.

(51) FIG. 3 illustrates a relationship between a hydraulic pressure fluctuation amount ΔP , which is the fluctuation amount of the braking hydraulic pressure P , and the drive power deviation ΔF . The relationship is one in which the drive power deviation ΔF is increased as the hydraulic pressure fluctuation amount ΔP is increased, and can be expressed as an approximate linear line L as illustrated in FIG. 3. A gradient of the approximate linear line L corresponds to the efficacy factor K .

(52) For example, the efficacy factor K can be calculated by the least-squares method. A description will be made on a calculation method by the least-squares method by using a relational equation (equation 1) to a relational equation (equation 3). The least-squares method calculates the efficacy factor K with which a sum-of-squares error J in the relational equation (equation 1) is minimized.

[Equation1]

$$(53) \quad J = \sum_{i=1}^n (F_i - (K P_i + \dots))^2 \quad (\text{equation1})$$

(54) The relational equation (equation 3) is obtained when the relational equation (equation 1) is solved in a manner to satisfy the relational equation (equation 2).

[Equation2]

[Equation3]

$$(55) \quad \frac{\partial J}{\partial K} = 0 \quad (\text{equation2}) \quad K = \frac{\sum_{i=1}^n (F_i P_i)}{\sum_{i=1}^n (P_i)^2} \quad (\text{equation3})$$

(56) The efficacy factor K can be calculated by the obtained relational equation (equation 3).

(57) The method for calculating the efficacy factor K is not limited to the calculation method by the least-squares method. For example, the efficacy factor K may be a value that is calculated as an arithmetic mean.

(58) <Micro-Braking Control>

(59) A description will be made on details of the micro-braking control that is executed by the braking control unit **12**. The micro-braking control refers to a series of control for actuating the friction braking system **70** by executing application processing and cancellation processing, which will be described below.

(60) FIG. 4 illustrates a processing flow that is executed by the braking control unit **12**. This processing routine is repeatedly executed at predetermined intervals.

(61) When this processing routine is started, first, in step **S101**, the braking control unit **12**

determines whether a start condition is satisfied. For example, the braking control unit **12** can determine that the start condition is satisfied in the case where both of the following condition A and the following condition B are satisfied. (Condition A) The calculation of vehicle weight M is completed. (Condition B) The drive power is being increased.

(62) If the start condition is not satisfied (**S101**: NO), the braking control unit **12** terminates this processing routine once.

(63) On the other hand, if the start condition is satisfied (**S101**: YES), the braking control unit **12** proceeds the processing to step **S102**.

(64) In step **S102**, the braking control unit **12** starts the application processing. In the application processing, the braking control unit **12** calculates the required braking force BP that corresponds to the generated drive power Fp. A description will be made on a relationship between the generated drive power Fp and the required braking force BP with reference to FIG. 5.

(65) FIG. 5 is an example of a map illustrating the relationship between the generated drive power Fp and the required braking force BP. The map illustrating such a relationship is stored in the braking control unit **12**.

(66) According to the relationship illustrating in FIG. 5, the required braking force BP is calculated within a range where the generated drive power Fp is equal to or higher than first drive power f1 and is equal to or lower than third drive power f3. In a range where the generated drive power Fp is equal to or higher than the first drive power f1 and is equal to or lower than second drive power f2, the larger required braking force BP is calculated as the generated drive power Fp is increased. The second drive power f2 corresponds to a specified threshold value of the generated drive power Fp. In a range where the generated drive power Fp is higher than the second drive power f2 and is equal to or lower than the third drive power f3, the smaller required braking force BP is calculated as the generated drive power Fp is increased. That is, the calculated required braking force BP becomes maximum when the generated drive power Fp is the second drive power f2. A maximum braking force bmax at the time when the generated drive power Fp is the second drive power f2 is set to a smaller value than the braking force that matches the second drive power f2. Here, matching means a state where the braking force cancels out the drive power and a movement amount of the vehicle **90** becomes “0” when it is assumed that the road surface is level. An example of the maximum braking force bmax is a magnitude that corresponds to 10% of the braking force matching the second drive power f2.

(67) In addition, the braking control unit **12** can set the front-rear distribution ratio a appropriately in the micro-braking control. In the case where the front-rear distribution ratio a is fixed to “0” or “1” during the execution of the micro-braking control, it is possible to select whether the braking force that is based on the calculated required braking force BP is applied to the front wheel **81** or the rear wheel **82**. As an example, the braking control unit **12** can alternately execute the micro-braking control for applying the braking force only to the front wheel **81** and the micro-braking control for applying the braking force only to the rear wheel **82**. Furthermore, the braking control unit **12** can also make the front-rear distribution ratio a fluctuate during the execution of the micro-braking control. The braking force can also be applied to the front wheel **81** and the rear wheel **82** by setting the front-rear distribution ratio a to a value that is larger than “0” and smaller than “1”.

(68) Referring back to FIG. 4, after starting the application processing and calculating the required braking force BP in step **S102**, the braking control unit **12** calculates the target value of the WC pressure on the basis of the required braking force BP and the conversion factor Kb. Next, the braking control unit **12** actuates the friction braking system **70** on the basis of the target value of the WC pressure. As a result, application of the braking force is started.

(69) Once the braking control unit **12** starts the application processing in step **S102**, the processing proceeds to step **S103**.

(70) In step **S103**, the braking control unit **12** determines whether a termination condition is satisfied. For example, in the case where a specified time has elapsed since the start of the

application processing, the braking control unit **12** determines that the termination condition is satisfied.

(71) In regard to the processing in step **S103**, the determination can be made as follows. Even in the case where the specified time has not elapsed since the start of the application processing, the braking control unit **12** can determine that the termination condition is satisfied. For example, in the case where the drive power starts reduced, the braking control unit **12** may determine that the termination condition is satisfied. Alternatively, for example, in the case where the operation of the braking operation member **79** by the driver intervenes, the braking control unit **12** can determine that the termination condition is satisfied. Furthermore, in the case where an operation speed of the braking operation member **79** is higher than a specified reduction operation determination value, the braking control unit **12** can determine that the termination condition is satisfied. The reduction operation determination value can be set as a value with which it is possible to determine that the driver requests rapid braking in the case where the operation speed is higher than the reduction operation determination value. Moreover, for example, in the case where an operation speed of the accelerator operation member by the driver is higher than a specified acceleration operation determination value, the braking control unit **12** can determine that the termination condition is satisfied. The acceleration operation determination value can be set as a value with which it is possible to determine that the driver requests rapid acceleration in the case where the operation speed is higher than the acceleration operation determination value. In the case where the rapid braking or the rapid acceleration is requested by the autonomous driving control device **30**, the braking control unit **12** may also determine that the termination condition is satisfied.

(72) If the termination condition is satisfied (**S103**: YES), the braking control unit **12** proceeds the processing to step **S104**. On the other hand, if the termination condition is not satisfied (**S103**: NO), the braking control unit **12** repeatedly executes the processing in step **S103**. The friction braking system **70** is actuated according to the application processing until the termination condition is satisfied.

(73) In step **S104**, the braking control unit **12** starts the cancellation processing. More specifically, the braking control unit **12** terminates the application processing and starts the cancellation processing. In the cancellation processing, the friction braking system **70** is actuated such that the braking force applied by the application processing is reduced to “0”. For example, the braking control unit **12** reduces the braking force at a predetermined reduction rate. The braking control unit **12** may bring the braking force into “0” within a predetermined time after the start of the cancellation processing. Once starting the cancellation processing, the braking control unit **12** terminates this processing routine.

(74) <Factor Estimation Processing>

(75) A description will be made on the factor estimation processing executed by the efficacy factor estimation unit **20**.

(76) FIG. **6** illustrates a processing flow that is executed by the efficacy factor estimation unit **20**. This processing routine is repeatedly executed at predetermined intervals.

(77) When this processing routine is started, first, in step **S201**, the efficacy factor estimation unit **20** determines whether the micro-braking control is being executed. If the micro-braking control is not being executed (**S201**: NO), the efficacy factor estimation unit **20** terminates this processing routine. On the other hand, if the micro-braking control is being executed (**S201**: YES), the efficacy factor estimation unit **20** proceeds the processing to step **S202**.

(78) In step **S202**, the efficacy factor estimation unit **20** determines whether the braking force is being increased. If the braking force that is applied to the wheel by the micro-braking control is being increased (**S202**: YES), the efficacy factor estimation unit **20** proceeds the processing to step **S203**.

(79) In step **S203**, the efficacy factor estimation unit **20** calculates the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP . The efficacy factor estimation unit **20**

temporarily stores the calculated values for use in the calculation of the efficacy factor K. A description will be made on the calculations of the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP with reference to FIG. 7.

(80) As illustrated in FIG. 7, the efficacy factor estimation unit **20** acquires the generated drive power F_p via the acquisition unit **13**. Furthermore, the efficacy factor estimation unit **20** acquires the vehicle weight M and the front-rear acceleration G_x via the acquisition unit **13**. The efficacy factor estimation unit **20** calculates the actual drive power F_x by multiplying the vehicle weight M by the front-rear acceleration G_x . The generated drive power F_p and the actual drive power F_x are input to the deviation calculation section **21** of the efficacy factor estimation unit **20**. The deviation calculation section **21** calculates the drive power deviation ΔF on the basis of the generated drive power F_p and the actual drive power F_x . At this time, the deviation calculation section **21** may calculate the drive power deviation ΔF by using a correction value calculated such that the drive power deviation ΔF at a time point when the micro-braking control is started becomes “0”.

(81) In addition, the efficacy factor estimation unit **20** acquires the braking hydraulic pressure P via the acquisition unit **13**. The braking hydraulic pressure P is input to the fluctuation pressure calculation section **22** of the efficacy factor estimation unit **20**. This braking hydraulic pressure P is the braking hydraulic pressure P corresponding to the wheel, to which the braking force is applied by the currently executed micro-braking control. The fluctuation pressure calculation section **22** compares the input braking hydraulic pressure P with the last braking hydraulic pressure P and calculates the hydraulic pressure fluctuation amount ΔP as the fluctuation amount of the braking hydraulic pressure P .

(82) Referring back to FIG. 6, after the efficacy factor estimation unit **20** calculates the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP in step S203, the efficacy factor estimation unit **20** proceeds the processing to step S204.

(83) On the other hand, if the braking force for the wheel that is subject to the micro-braking control is not being increased in the processing in step S202 (S202: NO), the efficacy factor estimation unit **20** proceeds the processing to step S204. That is, in the case where the braking force for the wheel that is subject to the micro-braking control remains constant, or in the case where the braking force is reduced, the efficacy factor estimation unit **20** does not execute the processing in step S203, and the processing proceeds to step S204.

(84) In step S204, the efficacy factor estimation unit **20** determines whether the application processing in the micro-braking control is terminated. If the application processing is terminated (S204: YES), the efficacy factor estimation unit **20** proceeds the processing to step S205.

(85) On the other hand, if the application processing is not terminated (S204: NO), the efficacy factor estimation unit **20** proceeds the processing to step S202. Accordingly, the processing from step S202 to step S204 is repeatedly executed until the application processing is terminated. At this time, when the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are calculated by the processing in step S203, the efficacy factor estimation unit **20** maintains the values of the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP that are calculated in the last time, and stores the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP that are newly calculated. In this way, the calculated values of the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are accumulated.

(86) In step S205, the efficacy factor estimation unit **20** estimates the efficacy factor K . The efficacy factor estimation unit **20** calculates the efficacy factor K on the basis of the drive power deviations ΔF and the hydraulic pressure fluctuation amounts ΔP , which are accumulated due to repeated execution of the processing in step S203.

(87) As illustrated in FIG. 7, the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are input to the output section **23** of the efficacy factor estimation unit **20**. The output section **23** uses the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP to calculate the efficacy factor K on the basis of the relational equation (equation 3). Referring back to

FIG. 6, after estimating the efficacy factor K in step S205, the efficacy factor estimation unit 20 terminates this processing routine.

(88) As illustrated in FIG. 7, the efficacy factor estimation unit 20 can store the calculated efficacy factor K in the storage unit 15. At this time, the storage unit 15 may store, in association with the efficacy factor K, the state amount of the vehicle 90 at the time when the efficacy factor K is calculated. For example, the storage unit 15 may store the temperature of the friction member 75 and the efficacy factor K in association with each other. The storage unit 15 may store the temperature of the friction member 75, the efficacy factor K, and the vehicle speed in association with each other.

(89) <Operation and Effects>

(90) A description will be made on operation and effects of this embodiment.

(91) In FIG. 8A, an example is illustrated in which the increased generated drive power F_p is reduced to "0" three times.

(92) In the example illustrated in FIGS. 8A and 8B, at timing t_{11} , it is determined that the start condition is satisfied, and the application processing in the micro-braking control is started (S102). Here, the braking force is applied only to the front wheel 81. Thus, as indicated by a solid line in FIG. 8B, the required front-wheel braking force BP_f starts increased at the timing t_{11} . At timing t_{12} , it is determined that the termination condition is satisfied, the application processing is terminated, and the cancellation processing is started (S104). Thus, at the timing t_{12} onward, the required front-wheel braking force BP_f is reduced.

(93) Here, for example, in some cases, the front-rear distribution ratio a is set in a manner to suppress pitching motion of the vehicle 90 when the vehicle 90 is decelerated due to braking. For this reason, in some cases, when it is attempted to apply the braking force for the estimation of the efficacy factor K during the deceleration, it is difficult to make adjustments such as applying the braking force only to either one of the front wheel 81 and the rear wheel 82 and changing the front-rear distribution ratio a . In the case where the front-rear distribution ratio a is changed during the deceleration in order to estimate the efficacy factor K, the pitching motion is possibly increased, and the occupant of the vehicle 90 possibly have a sense of discomfort.

(94) In order to handle this, the control device 10 starts the micro-braking control for executing the factor estimation processing when the drive power is increased. In this way, even when the braking force is generated on at least one wheel of the front wheel 81 and the rear wheel 82, it is possible to alleviate an impact of the generation of the braking force, such as the pitching motion of the vehicle 90 and the occupant of the vehicle 90 having the sense of discomfort.

(95) In addition, in the micro-braking control executed by the control device 10, it is ensured that the required braking force BP does not exceed the drive power. Thus, even when the micro-braking control is executed, the acceleration of the vehicle 90 is less likely to be suppressed.

(96) At timing t_{13} in FIGS. 8A and 8B, the generated drive power F_p is increased again, and the application processing is started. Here, the braking force is applied only to the rear wheel 82. Thus, as indicated by a broken line in FIG. 8B, the required rear-wheel braking force BP_r starts increased at the timing t_{13} . At timing t_{14} , it is determined that the termination condition is satisfied. Thus, at the timing t_{14} onward, the required rear-wheel braking force BP_r is reduced.

(97) At timing t_{15} onward, the application processing is started only for the front wheel 81, and the required front-wheel braking force BP_f is increased. At timing t_{16} onward, the required front-wheel braking force BP_f is reduced by the cancellation processing.

(98) In the example illustrated in FIGS. 8A and 8B, the required front-wheel braking force BP_f is increased in a period from the timing t_{11} to the timing t_{12} . Thus, the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are repeatedly calculated in such a period (S203). The hydraulic pressure fluctuation amount ΔP is calculated by using the braking hydraulic pressure P that corresponds to the pressing force for the front wheel 81. Then, when the application processing is terminated at the timing t_{12} , the efficacy factor K is estimated for the front wheel 81 (S205).

(99) Next, in a period from the timing **t13** to the timing **t14**, the required rear-wheel braking force **BPr** is increased. Thus, in such a period, the hydraulic pressure fluctuation amount ΔP is calculated by using the braking hydraulic pressure **P** that corresponds to the pressing force for the rear wheel **82**. When the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are repeatedly calculated, and the application processing is terminated at the timing **t14**, the efficacy factor **K** is estimated for the rear wheel **82**.

(100) In a period from the timing **t15** to the timing **t16**, similar to the period from the timing **t11** to the timing **t12**, the hydraulic pressure fluctuation amount ΔP is calculated by using the braking hydraulic pressure **P** that corresponds to the pressing force for the front wheel **81**. Then, the efficacy factor **K** is estimated for the front wheel **81**.

(101) In the micro-braking control executed by the control device **10**, the braking force is generated on at least one wheel of the front wheel **81** and the rear wheel **82**. Then, in the factor estimation processing, the pressing force corresponding to the wheel, to which the braking force is applied, is used to estimate the efficacy factor **K**. In this way, the efficacy factor **K** for the front wheel **81** and the efficacy factor **K** for the rear wheel **82** can be estimated distinctively.

(102) In the control device **10**, the drive power deviation ΔF , which is the deviation between the generated drive power F_p and the actual drive power F_x , is used to estimate the efficacy factor **K**. In regard to the vehicle **90** for which the electric motor is adopted as the power source **91**, calculation accuracy of the generated drive power F_p is likely to be higher than that in a case where the power source **91** is the internal combustion engine. For this reason, it is possible to accurately estimate the efficacy factor **K** for the vehicle **90** for which the electric motor is adopted as the power source **91**.

(103) In the control device **10**, the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP are calculated only when the braking force is being increased. That is, only the drive power deviation ΔF and the hydraulic pressure fluctuation amount ΔP at the time when the braking force is being increased are used to estimate the efficacy factor **K**. For this reason, a difference between the case where the braking force is being increased and the case where the braking force is being reduced does not affect the estimation of the efficacy factor **K**. Thus, a hysteresis characteristic of the friction braking system **70** can be ignored in the estimation of the efficacy factor **K**. According to the control device **10**, the efficacy factor **K** can be estimated accurately.

(104) According to the control device **10**, the conversion factor K_b can be corrected by using the efficacy factor **K**. In this way, the relationship between the pressing force and the braking force is made accurate, and it is thus possible to suppress deviation between the required braking force **BP** and the braking force that is actually applied to the wheel. A similar effect can be exerted when the efficacy factor **K** is used as the new conversion factor K_b .

(105) In the case where the vehicle **90** is operated by the driver, it can be expected that the driver performs the operation to eliminate the sense of discomfort associated with the deviation even when the deviation occurs between the required braking force **BP** and the braking force that is actually applied to the wheel. On the other hand, in the case where the autonomous travel is made, intervention by the driver is less likely to be expected. Accordingly, in the case where the autonomous travel is made, it is effective to accurately estimate the efficacy factor **K** like the control device **10** and use the efficacy factor **K** for the braking force control.

(106) In the case where a relationship between the pressing force for the front wheel **81** and the braking force significantly differs from a relationship between the pressing force for the rear wheel **82** and the braking force, a ratio of the actual front-wheel braking force that is applied according to the front-rear distribution ratio a possibly deviates from a ratio that is assumed on the basis of the front-rear distribution ratio a . In regard thereto, according to the control device **10**, the front-rear distribution ratio a can be adjusted on the basis of the difference between the efficacy factor **K** for the front wheel **81** and the efficacy factor **K** for the rear wheel **82**. In this way, it is possible to suppress the deviation of each of the ratio of the front-wheel braking force and the ratio of the rear-

wheel braking force from the desired ratio.

Modified Examples

(107) This embodiment can be implemented with the following modifications. This embodiment and the following modified examples can be implemented in combination with each other within the scope that is not technically contradictory.

(108) In the above embodiment, the example has been described in which the braking force for the front wheel **81** and the braking force for the rear wheel **82** are alternately applied in the case where the micro-braking control is executed repeatedly. The disclosure is not limited thereto. For example, the micro-braking control for applying the braking force to the front wheel **81** may be executed multiple times.

(109) In the above embodiment, the example has been described in which the braking force is applied to the front wheel **81** or the rear wheel **82** in the period from the start of the micro-braking control to the termination of the micro-braking control. Instead thereof, in the micro-braking control, the braking force can be applied to the front wheel **81** and the rear wheel **82**. A description will be made with reference to FIGS. **9A-9C**.

(110) In an example illustrated in FIGS. **9A-9C**, at timing **t21**, it is determined that the start condition is satisfied, and the application processing in the micro-braking control is started (**S102**). Here, first, the braking force is applied only to the front wheel **81**. As indicated by a solid line in FIG. **9B**, the required front-wheel braking force **BP_f** starts increased at the timing **t21**. The front-rear distribution ratio **a** at this time is “1” as illustrated in FIG. **9C**.

(111) As illustrated in FIG. **9C**, the front-rear distribution ratio **a** is gradually reduced from timing **t22**. As this result, as indicated by the solid line in FIG. **9B**, the required front-wheel braking force **BP_f** starts reduced. Furthermore, as indicated by a broken line in FIG. **9B**, the required rear-wheel braking force **BP_r** starts increased. At timing **t23**, the front-rear distribution ratio **a** is reduced to “0”.

(112) At the timing **t23**, it is determined that the termination condition is satisfied, the application processing is terminated, and the cancellation processing is started (**S104**). Thus, at the timing **t23** onward, the required rear-wheel braking force **BP_r** is reduced. At timing **t24**, the required rear-wheel braking force **BP_r** is “0”.

(113) In the example illustrated in FIGS. **9A-9C**, a period up to the timing **t23** is a period in which the required braking force **BP** is increased. During the execution of such micro-braking control, the factor estimation processing may be executed to estimate the efficacy factor **K**. In this case, the hydraulic pressure fluctuation amount ΔP for the front wheel **81** and the hydraulic pressure fluctuation amount ΔP for the rear wheel **82** may be calculated in consideration of the front-rear distribution ratio **a**.

(114) In the above embodiment, the example has been described in which the required braking force **BP** is calculated on the basis of the map illustrated in FIG. **5** in the micro-braking control. Instead thereof, an increase amount of the required braking force **BP** during the execution of the micro-braking control may be set.

(115) For example, the increase amount of the required braking force **BP** can be calculated by using the map in FIG. **5** in which the vertical axis represents the increase amount of the required braking force **BP**. That is, the increase amount of the required braking force **BP** can be set on the basis of a relationship between the generated drive power **F_p** and the increase amount of the required braking force **BP**. In this case, the increase in braking force is more likely to continue while the application processing is executed. That is, while the application processing executed, a period in which the braking force is kept constant and a period in which the braking force is reduced are less likely to occur. For this reason, it is possible to suppress extension of a period until the braking force that is applied by the micro-braking control becomes large. Thus, it is possible to reduce a time required to complete the estimation of the efficacy factor **K**. As a result, it is possible to reduce duration of the micro-braking control. By executing the micro-braking control in a short period, the

acceleration of the vehicle **90** is less likely to be interrupted.

(116) In addition, in the case where the generated drive power F_p is increased at specified increase rate F_{ps} or higher, the required braking force BP may be increased to the maximum braking force b_{max} at a preset increase rate. Just as described, in the micro-braking control, the increase rate of the required braking force BP can be set to the preset increase rate. For example, in the case where it is set that the increase rate of the required braking force BP fluctuates according to the increase rate of the generated drive power F_p , there is a possible delay in the increase of the required braking force BP or a possible sudden increase in the required braking force BP . In the case where the required braking force BP is increased rapidly, actuation noise may be increased in conjunction with the increase in the required braking force BP . On the contrary, in the case where the increase rate of the required braking force BP is set to the preset increase rate as in the above configuration, it is possible to suppress occurrence of the delay in the increase of the required braking force BP and the increase in the actuation noise.

(117) In the above embodiment, the condition A and the condition B are exemplified as the start conditions of the micro-braking control. However, the start condition is not limited thereto. The start condition may include that the vehicle speed is lower than a specified determination speed. The start condition may include that a change amount of the vehicle speed is smaller than a specified speed change amount. The start condition may include that the temperature of the friction member **75** is lower than a specified determination temperature.

(118) In the above embodiment, in the case where the braking force is increased as the processing in step **S202**, the processing in step **S203** is executed. The processing in step **S203** may be executed in the case where the braking force is equal to or larger than a specified determination braking force in addition to the increase in the braking force. In other words, the hydraulic pressure fluctuation amount ΔP and the drive power deviation ΔF in a period in which the braking force is smaller than the determination braking force during the execution of the micro-braking control may not be used to estimate the efficacy factor K .

(119) Even in the case where the braking force is increased during the execution of the micro-braking control, the hydraulic pressure fluctuation amount ΔP and the drive power deviation ΔF during turning may not be used to estimate the efficacy factor K when the vehicle **90** turns. In some cases, when the vehicle **90** turns, the actual drive power F_x is reduced due to a factor other than the braking force. By not using the hydraulic pressure fluctuation amount ΔP and the drive power deviation ΔF at the time when the actual drive power F_x is reduced due to the factor other than the braking force for the estimation, it is possible to eliminate the factor causing an error and to accurately estimate the efficacy factor K .

(120) Whether the vehicle **90** turns can be detected by an operation amount of a steering operation member, for example. In addition, turning of the vehicle **90** can be detected by a steering angle of a steering wheel. Turning of the vehicle **90** can also be detected on the basis of a steering command by the autonomous driving control device **30**.

(121) In the above embodiment, the hydraulic pressure fluctuation amount ΔP is calculated by using the braking hydraulic pressure P that is based on the detection value by the hydraulic pressure sensor **61**. The hydraulic pressure fluctuation amount ΔP can also be calculated on the basis of the target value of the pressing force, for example.

(122) The termination condition of the micro-braking control is not limited to that exemplified in the above embodiment. For example, even in the case where the specified time has not elapsed since the start of the application processing, it can be determined that the termination condition is satisfied when the vehicle **90** starts turning.

(123) In the above embodiment, the hydraulic braking system is exemplified as the friction braking system **70**. The friction braking system **70** is not limited to the hydraulic braking system, and a mechanical friction braking system, in which the friction member is pressed against the rotary body by mechanically transmitting a drive amount of the electric motor, may be adopted. In this case, the

magnitude of the pressing force corresponds to the drive amount of the electric motor. In such a configuration, the efficacy factor K can be estimated by using a fluctuation amount of the drive amount of the electric motor instead of the hydraulic pressure fluctuation amount ΔP in the above embodiment.

(124) In the above embodiment, the temperature of the friction member is estimated by the friction member temperature estimation unit **14**. The temperature of the friction member **75** may be acquired by a temperature sensor.

Claims

1. A control device that is applied to a vehicle having: a power source that generates drive power; and a friction braking system that applies a braking force to a wheel according to a pressing force to press a friction member against a rotary body integrally rotating with the wheel and can adjust the braking forces applied to a front wheel and a rear wheel of the wheels separately, the control device comprising: a braking control unit that controls the friction braking system; and an estimation unit that estimates an efficacy factor as a factor indicating a relationship between a magnitude of the pressing force and a magnitude of the braking force, wherein the braking control unit can start micro-braking control for generating the braking force on at least one wheel of the front wheel and the rear wheel in the case where the drive power is increased, and during execution of the micro-braking control, the estimation unit estimates the efficacy factor for the wheel, to which the braking force is applied by the micro-braking control, on the basis of a deviation between generated drive power that is based on a required value for the power source and actual drive power that actually acts on the vehicle, and the pressing force that corresponds to the wheel, to which the braking force is applied by the micro-braking control.
2. The control device according to claim 1, wherein in the micro-braking control, the braking control unit generates the braking force on either one wheel of the front wheel and the rear wheel.
3. The control device according to claim 1, wherein in the micro-braking control, the braking control unit generates the braking force on both wheels of the front wheel and the rear wheel on the basis of a front-rear distribution ratio indicating a ratio of the braking force applied to the front wheel to a total braking force that is a sum of the braking force applied to the front wheel and the braking force applied to the rear wheel.
4. The control device according to claim 3, wherein the braking control unit executes application processing to apply the braking force when the micro-braking control is started, executes cancellation processing to reduce the braking force to "0" after termination of the application processing, and terminates the micro-braking control, and when calculating the efficacy factor, the estimation unit uses the deviation between the generated drive power and the actual drive power in a period in which the braking force is increased.
5. The control device according to claim 4, wherein in the application processing, the braking control unit calculates, as the required braking force used for control of the friction braking system, a smaller value than the braking force that matches the generated drive power.
6. The control device according to claim 5, wherein in the application processing, the braking control unit calculates the required braking force according to the generated drive power, and calculates the larger required braking force as the generated drive power is increased within a range where the generated drive power is equal to or lower than a specified threshold value.
7. The control device according to claim 6, wherein in the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing in the case where a specified time has elapsed since the start of the application processing.
8. The control device according to claim 7, wherein in the case where the drive power starts reduced during the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing even when the specified time has not elapsed

since the start of the application processing.

9. The control device according to claim 8, wherein in the case where the vehicle starts turning during the execution of the micro-braking control, the estimation unit does not use the pressing force and the deviation between the generated drive power and the actual drive power in a period in which the vehicle turns for calculation of the efficacy factor.

10. The control device according to claim 5, wherein in the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing in the case where a specified time has elapsed since the start of the application processing.

11. The control device according to claim 4, wherein in the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing in the case where a specified time has elapsed since the start of the application processing.

12. The control device according to claim 3, wherein in the case where the vehicle starts turning during the execution of the micro-braking control, the estimation unit does not use the pressing force and the deviation between the generated drive power and the actual drive power in a period in which the vehicle turns for calculation of the efficacy factor.

13. The control device according to claim 2, wherein the braking control unit executes application processing to apply the braking force when the micro-braking control is started, executes cancellation processing to reduce the braking force to “0” after termination of the application processing, and terminates the micro-braking control, and when calculating the efficacy factor, the estimation unit uses the deviation between the generated drive power and the actual drive power in a period in which the braking force is increased.

14. The control device according to claim 13, wherein in the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing in the case where a specified time has elapsed since the start of the application processing.

15. The control device according to claim 2, wherein in the case where the vehicle starts turning during the execution of the micro-braking control, the estimation unit does not use the pressing force and the deviation between the generated drive power and the actual drive power in a period in which the vehicle turns for calculation of the efficacy factor.

16. The control device according to claim 1, wherein the braking control unit executes application processing to apply the braking force when the micro-braking control is started, executes cancellation processing to reduce the braking force to “0” after termination of the application processing, and terminates the micro-braking control, and when calculating the efficacy factor, the estimation unit uses the deviation between the generated drive power and the actual drive power in a period in which the braking force is increased.

17. The control device according to claim 16, wherein in the micro-braking control, the braking control unit terminates the application processing and starts the cancellation processing in the case where a specified time has elapsed since the start of the application processing.

18. The control device according to claim 1, wherein in the case where the vehicle starts turning during the execution of the micro-braking control, the estimation unit does not use the pressing force and the deviation between the generated drive power and the actual drive power in a period in which the vehicle turns for calculation of the efficacy factor.
